
CIPHER: A Benchmark for Captcha Instruction Parsing with Hidden Extraction and Reasoning

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Abstract

We introduce **CIPHER** (Captcha Instruction Parsing with **H**idden **E**xtraction and **R**easoning), a procedurally generated benchmark for evaluating vision-language models (VLMs) on video-editing captcha tasks. In CIPHER, an agent receives a short video clip together with natural-language editing instructions (e.g. “reverse the video, then flip horizontally”) and must (i) parse the instructions into a sequence of edit operations, (ii) execute those operations via `ffmpeg`, and (iii) extract a short alphanumeric key that becomes visible in the correctly edited result. The benchmark spans five difficulty levels that independently vary key visibility, operation count, and instruction ambiguity, yielding 250 procedurally generated samples backed by diverse Kinetics-400 background clips. We evaluate OCR baselines, an agentic MCP pipeline, and frontier VLMs (Claude Opus/Sonnet/Haiku), finding that S3 key extraction in isolation is near-trivial for modern VLMs (Opus/Sonnet: 100% EM) but OCR baselines struggle ($\sim 25\%$ EM). The compound difficulty of CIPHER lies in the full pipeline: temporal key localization, correct operation sequencing, and tool execution create a challenging integration problem where no single stage dominates.

1 Introduction

Captchas have evolved from distorted text images to dynamic, multi-step challenges designed to resist automation while remaining trivially solvable by humans. A recent class of video-editing captchas presents users with a short clip and a natural-language instruction such as “reverse the video, then flip it horizontally.” Imagine a user trying to log into a secure system, only to be confronted with a 15-second chaotic video clip and the instruction: “The video has been horizontally mirrored, then sped up 2x. Undo these transformations.” A human with basic video editing software can quickly apply a horizontal flip and halve the playback speed, bringing a briefly flashing, faint alphanumeric key (e.g., ‘K9M2P’) into clear, readable view. This seemingly simple test effectively filters out bots while remaining manageable for human problem-solvers. An automated agent must parse the instruction, execute the correct sequence of video transformations, and extract the now-visible key—a compound task spanning language understanding, procedural execution, and visual perception.

Despite the proliferation of video-capable VLMs [OpenAI, 2023, Google, 2023, Wang et al., 2024], no benchmark systematically tests this compound capability. While current video understanding benchmarks such as Video-MME [Fu et al., 2024] and LongVideoBench [Wu et al., 2024] assess passive temporal reasoning and question answering, they do not require active, sequential manipulation of the visual stream based on language. Conversely, video generation and editing benchmarks like VBench [Huang et al., 2023] evaluate the perceptual quality of generative models rather than the precise execution of rigid transformational rules necessary for agentic problem-solving. Neither paradigm captures the tight coupling of instruction-following, procedural tool use, and visual extraction required here. CIPHER fills this gap.

Contributions.

Preprint.

1. **Task formulation.** We define the video-editing captcha solving task as a three-stage pipeline: instruction parsing (S1), ffmpeg execution (S2), and key extraction (S3), enabling fine-grained per-stage diagnostics.
2. **Benchmark.** CIPHER provides 250 procedurally generated samples across five difficulty levels (L1–L5), varying key visibility, operation count (1–4 ops), and instruction ambiguity (exact to compositional). Backgrounds are drawn from Kinetics-400 to ensure visual diversity.
3. **Metrics.** We propose *Pipeline-F1*, a weighted combination of per-stage scores ($0.25 \cdot S1 + 0.25 \cdot S2 + 0.50 \cdot S3$), alongside strict exact match (EM) and character-level F1.
4. **Baselines.** We benchmark OCR-only, an agentic MCP pipeline, and frontier VLMs (Claude Opus/Sonnet/Haiku, GPT-5.4), revealing that S3 key extraction is near-trivial for VLMs in isolation but the full agentic pipeline—parsing, execution, temporal key localization—remains a challenging integration problem.

2 Task Definition

Setting. Let $\mathcal{V}$ be the space of all video clips and $\mathcal{L}$ be the space of natural language strings. Formally, the problem is modeled as finding a function $f : \mathcal{V} \times \mathcal{L} \rightarrow \Sigma^7$ that maps a scrambled video v' and a natural-language instruction $\mathcal{I}$ to a predicted alphanumeric key $\hat{k}$. The original video v contains a latent key $k^* \in \Sigma^7$ embedded as text during a specific temporal window $[t_s, t_e]$. A sequence of video transformation operations $\mathbf{o} = (o_1, \dots, o_n)$ is applied to v to yield the input $v' = o_n \circ \dots \circ o_1(v)$. The optimal agent must invert this process: inferring the correct inverse operations $\mathbf{o}^{-1}$ from $\mathcal{I}$, executing them to restore v , and extracting k^* . The key k^* is drawn from unambiguous alphanumerics ($\Sigma = \text{A-Z} \cup 2-9 \setminus \{\text{O}, 0, \text{I}, 1\}$). $\mathcal{I}$ is generated from $\mathbf{o}$ via paraphrase templates.

Three-stage pipeline. This task is naturally highly composite. An end-to-end failure (i.e., $\hat{k} \neq k^*$) could stem from a hallucinated operation, a malformed execution command, or simply failing to perceive the key against a cluttered background. By enforcing a three-stage pipeline—Instruction Parsing (S1), Execution (S2), and Key Extraction (S3)—CIPHER isolates these failure modes. It allows researchers to pinpoint whether a model’s bottleneck lies in its linguistic reasoning and tool-use (S1, S2) or in its foundational spatio-temporal perception capabilities (S3), preventing monolithic metrics from masking specific architectural deficiencies.

S1 Parse: $\mathcal{I} \rightarrow \hat{\mathbf{o}}$. Score: sequence-aware op F1.

S2 Execute: run `ffmpeg`($\hat{\mathbf{o}}$). Score: binary validity.

S3 Extract: read $\hat{k}$ from edited video. Score: CharF1 + EM.

Operations. Six composable ffmpeg ops: `reverse`, `hflip`, `vflip`, `rotate90`, `rotate270`, `speed2x`.

3 The CIPHER Benchmark

3.1 Difficulty levels

Table 1: CIPHER difficulty levels (50 samples each, $N=250$ total).

Level	Key vis.	Ops	NL amb.	Distr.	Primary challenge
L1	1 (large)	1	1	0	Single-op baseline
L2	1 (large)	2	1	0	Two-op chaining
L3	2 (medium)	2	2	1	Paraphrase + distractor
L4	2 (medium)	3	2	2	Three-op + distractors
L5	3 (small)	4	3	3	Max complexity

Key visibility varies font size (13%→7% of frame height), alpha (255→200), and position randomization (centered→random). The key is always rendered in bright yellow (RGB 255,255,0) with a black drop shadow; distractors use non-yellow colors (cyan, green, magenta) and are always smaller and dimmer.

NL ambiguity levels: (1) exact canonical phrasing; (2) paraphrase from a pool of 4–6 templates per operation; (3) compositional with “undo” framing and embedded temporal cues. All phrases are designed to be parseable by regex for S1 baseline scoring.

Operation Vocabulary. The operational vocabulary consists of six composable transformations: `reverse`, `hflip`, `vflip`, `rotate90`, `rotate270`, and `speed2x`. These operations are visually distinct, non-destructive, and strictly composable via standard `ffmpeg` filters. They demand different types of reasoning: spatial (`hflip`, `vflip`, `rotate`), and temporal (`reverse`, `speed2x`), providing a comprehensive testbed for multimodal execution without requiring complex parameter tuning from the model.

3.2 Procedural generation and Dataset Integrity

Samples are fully deterministic from $(seed, level)$. Background clips are drawn uniformly from a Kinetics-400 pool (diverse human activities, minimal on-screen text). Test split: seeds 0–49. Challenge split: seeds 50–99.

A persistent challenge in evaluating large language and vision models is the risk of data contamination, where static benchmark data inadvertently leaks into the pre-training corpus. CIPHER mitigates this risk entirely through programmatic dataset synthesis. The benchmark is procedurally generated from a tuple of $(seed, level)$, which deterministically selects the background clip, the randomly generated 7-character key, the sequence of operations, and the natural language instruction template. Because samples are dynamically constructed on-demand using disjoint background video pools, researchers can regenerate the benchmark with novel seeds, guaranteeing an uncontaminated evaluation environment and an effectively infinite supply of test instances.

3.3 Difficulty Calibration via GEPA

A benchmark that saturates (100% EM) or collapses (0% EM) across all models conveys no gradient. To locate a generator configuration that produces partial credit for frontier VLMs, we adopt a GEPA-style [Khattab et al., 2025] reflective evolutionary loop over the image-mode generator. The generator is specified as an S-expression DSL over five tunable numerical parameters: key alpha α , font size as a fraction of frame height, number and chromatic composition of distractors, Gaussian noise σ , and Gaussian blur radius. Each *program* corresponds to a fixed instantiation of these parameters; *fitness* is defined as $-|EM - 0.03| - 0.5 \cdot \text{miss rate}$, targeting a 3% sweet spot at which models read but fail on individual characters.

We evaluated four rounds of five programs each, with four samples per program (eight in the last round), using Gemini Flash as the calibration model. Round 1 performed a wide parameter sweep; subsequent rounds progressively narrowed the region around the fitness peak. Round 3 produced a program ($\alpha=145$, font 4.8%, three distractors with two yellow, noise $\sigma=4$, blur 0.25) yielding 0% EM with predictions of the correct length—a *near-miss* regime where characters such as Q/O, 8/B, X/K, 6/G are consistently confused. This regime is what we use as the default L3–L5 extraction condition: it is solvable but not trivial for Gemini-class readers, while remaining entirely uninformative (0% EM) for non-VLM baselines.

3.4 Metrics

EM. $\frac{1}{N} \sum_i \mathbf{1}[\hat{k}_i = k_i^*]$.

CharF1. Character-level precision/recall F1 (partial credit).

PipeF1.

$$\text{PipeF1} = 0.25 \cdot \overline{S1} + 0.25 \cdot \overline{S2} + 0.50 \cdot \overline{\text{CharF1}} \quad (1)$$

S3 receives double weight as the terminal objective; S1/S2 serve as diagnostic leading indicators.

4 Experiments

4.1 Experimental Setup

Inference modes. We evaluate models in two settings: (1) **S3-only**: models receive a ground-truth corrected frame containing the key and must read the yellow 7-character code (tests pure visual extraction); (2) **Full pipeline**: models receive the scrambled .mp4 video and NL instructions with Bash access, and must independently parse instructions (S1), execute ffmpeg corrections (S2), locate the key frame, and extract the key (S3).

Agentic evaluation. For full-pipeline evaluation, VLM agents are given Bash access and must decide their own video processing strategy (ffmpeg flags, frame sampling, temporal search). No ground-truth metadata (operations, key, timing) is provided. This design ensures fairness across models with different video processing capabilities.

Frame sampling. In S3-only evaluation, key frames are identified programmatically via edge-aware yellow pixel detection (Sobel gradient > 30 in yellow regions, horizontal aspect ratio > 2.5) to distinguish text from yellow background elements.

4.2 Models

We benchmark seven configurations spanning OCR baselines, API-first VLMs, and agentic pipelines:

OCR-only Pure extraction baseline. Uses easyocr on sampled key frames; no language understanding or color filtering.

Yellow-OCR Color-filtered OCR baseline. Isolates yellow pixels via HSV masking (hue 20–40, saturation > 100 , value > 150) before running easyocr. Tests whether color-aware preprocessing can substitute for VLM reasoning.

MCP Skill Agentic pipeline baseline (`solve_captcha.py`): regex-based instruction parsing (S1), ffmpeg execution (S2), and a multi-method extraction chain (S3): QR $\rightarrow$ OCR $\rightarrow$ VLM $\rightarrow$ audio morse $\rightarrow$ LSB steganography.

Claude Haiku (claude-haiku-4-5) Compact API-first VLM from Anthropic. Optimized for latency and cost.

Claude Sonnet (claude-sonnet-4-6) Mid-tier Anthropic VLM. Balances capability and throughput.

Claude Opus (claude-opus-4-6) Frontier Anthropic VLM. Establishes the upper bound for single-provider scaling.

GPT-5.4 (codex detailed) OpenAI frontier model with agentic tool use. Evaluated via Swarm orchestration with full Bash access.

4.3 Main Results

Table 2: S3 key extraction from ground-truth corrected frames ($n=5$ per level). Best per column in **bold**.

Model	L1	L2	L3	L4	Total
OCR-only	20%	25%	30%	25%	25%
Yellow-OCR	33%	—	—	—	—
MCP Skill	20%	25%	30%	25%	25%
Haiku	100%	60%	40%	100%	75%
Sonnet	100%	100%	100%	100%	100%
Opus	100%	100%	100%	100%	100%

Overview. Table 2 reports S3 key extraction accuracy in isolation: each model receives a ground-truth corrected frame containing the yellow key and must read it. Frontier VLMs (Opus, Sonnet) achieve perfect accuracy, while Haiku drops to 75% on frames where the key is smaller and randomly positioned (visibility level 2). OCR baselines perform poorly ($\sim 25\%$) due to background text

Table 3: Full pipeline results (video $\rightarrow$ key). Agents receive scrambled .mp4 and NL instructions with Bash access. TODO: complete after full-pipeline eval.

Model	L1 EM	L2 EM	L3 EM	L4 EM	EM (all)	PipeF1
MCP Skill	–	–	–	–	–	–
Haiku	–	–	–	–	–	–
Sonnet	–	–	–	–	–	–
Opus	–	–	–	–	–	–
GPT-5.4	–	–	–	–	–	–

interference from Big Buck Bunny credits and scene elements. Crucially, S3 in isolation is *not* the binding bottleneck for VLMs—the compound difficulty of CIPHER lies in the full pipeline, where agents must also identify the correct temporal window containing the key.

4.4 Per-Level Breakdown

To isolate the effect of difficulty progression, we analyze exact match separately by level. Figure ?? shows EM curves for each model across L1–L5. We expect monotonic (or near-monotonic) EM decrease with level difficulty due to compounding effects of low key visibility, increased operation count, and instruction ambiguity. Comparing levels reveals which capability gaps are inherent to model scale versus which stem from brittleness in specific reasoning stages (e.g., rotating operations vs. speed-related paraphrase). This breakdown also pinpoints where the benchmark meaningfully separates models: models that saturate on L1–L2 do not benefit from the data collected at L3–L5.

5 Analysis

5.1 Stage bottleneck analysis

We decompose failures by stage to identify where the pipeline breaks down. Figure ?? shows per-level S1/S2/S3 scores for each model.

S1 (parsing) saturates quickly across models: regex matching achieves high parse F1 on L1–L3. L5 compositional instructions require VLM-based parsing.

S2 (execution) failures are rare on L1–L2 but increase with op count (L4–L5), primarily due to ffmpeg filter chain errors when ops compose non-trivially (e.g., `reverse` requires a separate pass before other filters).

S3 (extraction) is *not* the binding bottleneck for VLMs. In isolation (given the correct frame), Opus and Sonnet achieve 100% EM across all levels. Haiku drops to 75% on smaller, randomly-positioned keys (visibility level 2). OCR baselines ($\sim 25\%$ EM) struggle with background text from video credits and scene elements. The true difficulty lies in **temporal key localization**: the key appears only during a 2.5–4.5s window within the video, and agents must identify the correct frames before extraction.

5.2 NL ambiguity ablation

Holding key visibility and op count fixed (L2 vs. L3), we isolate the effect of instruction paraphrase. Parse F1 drops by Δ on paraphrase vs. exact phrasing, with most errors on speed and rotation ops where synonym coverage is lowest.

5.3 Background influence

All backgrounds are drawn from Kinetics-400 (natural track). As a diagnostic, we rerun OCR-only on a 50-sample clean-background subset (BBB clips). CharF1 improves by Δ , confirming that natural backgrounds introduce measurable OCR distraction even at high key visibility.

5.4 Character-level confusion in the near-miss regime

Under the calibrated extraction condition (Section 3.3), Gemini Flash outputs predictions of the correct key length but makes single-character substitutions that are *consistent* with the visual geometry of the rendered glyphs, not random guessing. Representative errors from Round 3 `prog_3`:

TQ573CYX $\rightarrow$ T0573CYK, D3JMQKWH $\rightarrow$ D3JMOKWH, 8ER7C6LN $\rightarrow$ BER7CGLN.

The dominant substitution classes are $\{Q \leftrightarrow O, Q \leftrightarrow 0\}$, $\{8 \leftrightarrow B\}$, and $\{X \leftrightarrow K, 6 \leftrightarrow G\}$. These pairs share identical bounding-box silhouettes once the noise-plus-blur filter discards interior strokes, suggesting that the binding constraint at this scale is low-frequency glyph shape rather than instruction understanding. Invalidated Round 4 (fallback to Claude after Gemini was unreachable) produced $<12\%$ char-accuracy and zero positionally correct characters, illustrating that outside the near-miss regime the same images elicit qualitatively different failure modes (fabrication vs. misreading).

6 Conclusion

We introduced CIPHER, a procedurally generated benchmark for video-editing captcha solving that stress-tests VLMs on a compound task: parsing natural-language edit instructions, executing an ffmpeg transformation chain, and extracting a hidden key from the result. The three-stage pipeline structure and PipeF1 metric enable fine-grained diagnosis of where agent capability breaks down.

Our experiments reveal that frontier VLMs solve S3 key extraction near-perfectly in isolation (Opus/Sonnet: 100% EM), while OCR baselines struggle ($\sim 25\%$). The compound difficulty of CIPHER lies in the full agentic pipeline: temporal key localization, correct operation sequencing, and tool execution create a challenging integration problem that no evaluated model solves reliably across all levels.

Limitations. CIPHER focuses on seven deterministic ffmpeg operations; generative edits (e.g., inpainting, style transfer) and audio-only keys are out of scope. Human baselines are not included in this version.

Future work. L6 (QR/morse key encoding), multi-agent solving, and adversarial instruction generation are natural extensions. The procedural generator enables zero-cost dataset refresh, making CIPHER suitable for ongoing leaderboard tracking.

Reproducibility. All code and generation scripts are released at <https://github.com/Fr0do/cipher>. The test manifest (seeds 0–49) is pinned at commit TODO.

Acknowledgements

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